

PJM Wind Farm Generation Simulation

High-Accuracy Wind Power Modeling Using SAM and NASA POWER Data

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Executive Summary

This project developed a high-accuracy calibrated wind generation model for PJM wind farms using the System Advisor Model (SAM) and NASA POWER weather data. The calibrated model achieves 99.1% correlation ($R^2 = 0.991$), with 68% of farms predicted within $\pm 5\%$ accuracy for 2023 data. The model also shows promise in being applied generally to past and future years, with moderate-to-strong correlation ($R^2 = 0.89$) and 38% of farms being predicted within $\pm 5\%$ accuracy for 2022 data.

Key Results:

Metric (2023)	Value
Total Farms Simulated	106
Farm-Level Correlation (R^2)	0.991
RMSE	23,716 MWh
Farms within $\pm 5\%$ accuracy	68% (72 farms)
Farms within $\pm 10\%$ accuracy	84% (89 farms)

Metric (2022)	Value
Total Farms Simulated	99
Farm-Level Correlation (R^2)	0.89
RMSE	105,960 MWh
Farms within $\pm 5\%$ accuracy	38% (38 farms)
Farms within $\pm 10\%$ accuracy	47% (47 farms)

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1. Project Overview

1.1 Objective

Develop a high-accuracy wind generation model to simulate hourly electricity production for all wind farms in the PJM interconnection region.

1.2 Scope

The project encompasses:

- 106 wind farms across 8 states (IL, IN, OH, PA, WV, MD, VA, NC)
- Total installed capacity of 10,807.5 MW
- Hourly generation simulation for 8,760 hours (2023)
- Site-specific turbine modeling using EIA-860 data
- NASA POWER weather data integration
- Advanced three-layer calibration approach

To construct this, the model will be optimized around relevant data for 2023. Afterward, a 2022 analysis is presented as an out-of-sample validation test.

1.3 Tools and Software

Tool	Purpose
SAM (System Advisor Model)	NREL wind power simulation engine
Python 3.x	Data processing and automation
NASA POWER API	Hourly weather data (0.5° resolution)
EIA-860	Turbine specifications database
PySAM	Python interface to SAM

2. Literature Review

2.1 Wind Plant Validation Data

Several recent studies rely on public datasets such as EIA Form 860 for wind plant metadata and EIA Form 923 for monthly or annual generation validation.[1,3] These datasets are useful because they provide standardized information about plant capacity, location, and observed generation. This is relevant

to the present project because the model uses plant-level information to simulate wind output and compares the results against reported generation values.

2.2 Meteorological Data Sources for Wind Modeling

A major challenge in wind generation modeling is choosing an appropriate weather dataset. Prior studies have used sources such as ERA5, MERRA2, HRRR, and NASA POWER to estimate wind speed, wind direction, temperature, pressure, and air density.[2] These sources differ in spatial resolution, temporal resolution, and bias, which can lead to different simulated generation results. One study notes that errors can occur when wind speeds are not available at the exact turbine hub height, requiring extrapolation.[5] This is directly connected to this project because NASA POWER provides wind speeds at limited heights, so the model must estimate wind speeds at turbine hub height using a wind shear assumption.

2.3 Power Curve Modeling and Turbine Performance

Other studies focus on the conversion from wind speed to electrical power. Although theoretical turbine power curves are useful, actual turbine output may deviate from manufacturer curves due to air density, turbine condition, wake effects, control behavior, and local wind conditions. Some researchers use historical turbine output and weather data to construct empirical power curves using parametric models or machine learning approaches.[3, 4] This suggests that using generic or manufacturer power curves may introduce error, especially when exact turbine models or hub heights are unavailable.

2.4 Air Density, Wind Shear, and Wake Effects

Recent work also highlights the importance of physical factors that are sometimes simplified in wind generation models. Assuming constant air density or a fixed wind shear exponent can lead to significant errors. Similarly, wake effects from upstream turbines can reduce wind speeds for downstream turbines, causing models to overestimate available wind resources.[5] These issues are relevant to this project because the current model uses simplified assumptions, including wind speed extrapolation and plant-level equivalent turbine scaling.

2.5 Relevance to This Project

Together, these sources show that wind generation modeling is usually built from four main components: plant metadata, meteorological data, power conversion, and validation against observed generation. This project follows this same general structure, using PJM wind plant data, NASA POWER weather data, turbine power curves, and EIA generation data. However, the literature also suggests several possible sources of error, including weather dataset bias, hub-height uncertainty, simplified wind shear assumptions, air density effects, and unavailable turbine-specific information. These limitations provide context for evaluating the accuracy of the model and suggest a framework for how it should be constructed.

3. Data Sources

3.1 Wind Farm Data

Wind farm operational data was obtained from PJM generation reports for 2023, including plant codes, names, locations, capacities, and actual annual generation.

Data Filtering Process:

Category	Count
Initial wind plants in PJM	114
Excluded: Zero/blank generation	8
Final plants included	106

3.2 Turbine Specifications (EIA-860)

The U.S. Energy Information Administration Form EIA-860 provided detailed turbine-level data for 94 of 106 farms (89%), including:

- Turbine manufacturer and model
- Number of turbines per farm
- Hub height (converted from feet to meters)
- Rotor diameter (derived from turbine model)
- Nameplate capacity per turbine

Handling Missing EIA-860 Data:

For farms with incomplete or missing EIA-860 data, the following defaults were applied:

Parameter	Default Value	Rationale
Number of turbines	Estimated from capacity / 2.0 MW	Assumes typical 2 MW turbines
Hub height	90 meters	Industry standard for modern turbines
Rotor diameter	110 meters	Typical for 2-3 MW class turbines
Manufacturer/Model	Capacity-based match	Selected from SAM library by MW rating

3.3 Weather Data (NASA POWER)

NASA POWER (Prediction of Worldwide Energy Resources) provides satellite-derived and reanalysis weather data at $0.5^\circ \times 0.5^\circ$ spatial resolution and hourly temporal resolution. Variables retrieved: wind speed at 10m and 50m, air temperature, and surface pressure.

4. Methodology

4.1 Turbine Selection Process

For each wind farm, turbines were selected from SAM's library of 300+ turbine power curves using a two-tier matching system:

Tier 1: Manufacturer/Model Matching (94 farms)

- Exact or fuzzy string matching between EIA-860 turbine data and SAM library. Examples: "Vestas V110-2.0", "Gamesa G87 2.0 MW", "GE 1.5 xle".

Tier 2: Capacity-Based Matching (12 farms)

- For farms without manufacturer/model data in EIA-860, turbines with rated capacity closest to $\text{farm_capacity} / \text{number_of_turbines}$ were selected. For example, a 100 MW farm with 50 turbines (2.0 MW each) would be matched to a generic 2.0 MW turbine from SAM's library, using actual hub height and rotor diameter if available, or defaults (90m hub, 110m rotor) if not.

4.2 Initial Model (Uncalibrated)

Baseline model used uniform loss factors across all farms:

Loss Category	Value	Description
Balance of Plant (BOP)	10%	Wake losses, array losses
Grid/Electrical	10%	Transmission, transformer losses
Turbine Availability	12%	Downtime, degradation, curtailment
Total System Loss	28.3%	Combined effect

Uncalibrated Results:

Metric	Value
R ² Correlation	0.920
Overall Bias	+1.8%
RMSE	71,939 MWh
Farms within ±5%	16%
Farms with >20% error	36%

5. 2023 Model Calibration

To improve accuracy from $R^2 = 0.920$ to 0.991, a three-layer calibration approach was developed:

5.1 Layer 1: Terrain-Based Adjustments

Balance of Plant (BOP) losses were adjusted based on state-level terrain characteristics. Base BOP loss is 10%, which is multiplied by a terrain factor:

State	Adjustment	Actual BOP Loss	Terrain
Pennsylvania (PA)	+10%	11.0%	Mountainous terrain
West Virginia (WV)	+15%	11.5%	Very mountainous
Illinois (IL)	-5%	9.5%	Flat farmland
Indiana (IN)	-5%	9.5%	Flat farmland
Ohio (OH)	0%	10.0%	Rolling hills
Virginia (VA)	-10%	9.0%	Offshore wind

Note: Grid losses (10%) and base turbine availability losses (12%) remain constant across all states. Only BOP losses vary by terrain.

Scientific Basis:

Terrain-based adjustments are supported by extensive research on wake losses and terrain effects in wind farms:

- Complex terrain (mountains) increases turbulence intensity and wake losses by 10-20% (Brower, 2012)
- Flat terrain reduces wake interference, improving array efficiency by 5-10% (Barthelmie et al., 2009)
- Offshore wind farms experience 10-15% lower wake losses due to reduced surface roughness (Barthelmie et al., 2009)
- Mountainous sites show 15-25% higher energy losses compared to flat sites at same wind speeds (Katic et al., 1986)

5.2 Layer 2: Curtailment Modeling

Large wind farms (>150 MW) experience higher curtailment due to grid constraints and transmission limitations. Formula: $\text{curtailment_loss} = \min(5\%, (\text{capacity_MW} - 150) / 50)$

Scientific Basis:

Curtailment modeling is based on empirical observations of large wind farm operations:

- Large wind farms (>150 MW) experience 2-5% higher curtailment than smaller farms (Wiser et al., 2016)
- Grid integration constraints increase with farm size, particularly in congested transmission areas (Clifton et al., 2016)
- PJM region shows higher curtailment rates for large farms during low-demand periods (PJM, 2023)

5.3 Layer 3: Individual Farm Calibration

For each farm, losses were individually calibrated to match actual 2023 generation while maintaining physical realism (losses clamped to 0-25% per category).

Scientific Basis:

Individual calibration accounts for site-specific factors not captured by generic models:

- Wind farm performance varies by 10-30% due to site-specific factors (micro-siting, local obstacles, maintenance) (Staffell & Green, 2014)
- Turbine degradation and availability vary significantly between farms (5-15% annual variation) (Staffell & Green, 2014)

- Atmospheric stability and turbulence intensity affect power production by 10-20% (St. Martin et al., 2016)
- Model calibration to actual data improves accuracy from $R^2=0.85-0.92$ to $R^2>0.95$ (Clifton et al., 2016)

6. 2023 Calibrated Results and Validation

6.1 Performance Improvement

Metric	Before	After	Change
R² Correlation	0.920	0.991	+7.7%
RMSE (MWh)	71,939	23,716	-67%
Mean Absolute Error	45,340	12,869	-72%
Overall Bias	+1.8%	+1.3%	Better
Farms within $\pm 5\%$	16%	68%	+325%
Farms within $\pm 10\%$	34%	84%	+147%
Farms with $>20\%$ error	36%	11%	-69%

6.2 Overall Statistics

Statistic	Value
Total Farms	106
Total Capacity	10,807.5 MW
Actual Generation	27,920,675 MWh (27.9 TWh)

Simulated Generation	28,277,443 MWh (28.3 TWh)
Bias	+356,768 MWh (+1.3%)
Average Capacity Factor	29.9%
R²	0.991
RMSE	23,716 MWh

6.3 Accuracy Distribution

Category	Count	Percentage
Excellent (<5% error)	72 farms	67.9%
Good (5-10% error)	17 farms	16.0%
Fair (10-20% error)	5 farms	4.7%
Poor (>20% error)	12 farms	11.3%

6.4 Visualization and Analysis Plots

Comprehensive visualizations were generated to support the analysis:

- Farm-by-farm comparison: SAM vs Actual generation scatter plots with $R^2 = 0.991$
- Bias distribution: Histogram showing model accuracy across all 106 farms
- State-level performance: Terrain-adjusted bias by state (PA, WV, IL, IN, OH, VA)
- Time series validation: Monthly generation profiles comparing SAM vs actual
- Calibration layers: Before/after comparison for terrain, curtailment, and individual adjustments
- Top performers: Charts highlighting best and worst performing farms
- Turbine matching: Distribution of turbine types and matching success rates
- Capacity factor analysis: Modeled vs actual capacity factors by farm

7. 2023 Farm-Level Analysis

7.1 Best Performing Farms (Top 5)

Farm Name	Capacity	Bias
Blue Creek Wind (OH)	302.0 MW	+0.0%
Headwaters Wind Farm (IN)	200.0 MW	+0.2%
Big Sky Wind (IL)	239.4 MW	+0.2%
Streator Cayuga Ridge (IL)	300.0 MW	-0.2%
Meadow Lake V (IN)	100.0 MW	+0.4%

7.2 Challenging Farms

Twelve farms still exhibit >20% error after calibration, primarily due to small farm size, old turbines, data quality issues, or extreme terrain.

8. 2022 Validation

The calibrated model was applied to wind speed and wind farm data from 2022 to test its generality. For this test, the exact same algorithm was used with the allowance for larger hub heights to have their speeds be extrapolated from wind readings at 50 meters instead of 10 meters. This shift should only make the wind readings more accurate and less of an estimate.

8.1 Applicable Farms

Statistic	Value
Total Farms	106
Excluded Farms	7
Remaining Farms	99

8.2 Overall Statistics

Metric	Value
R² Correlation	0.8897
RMSE (MWh)	105,959.51
Mean Absolute Error	56,983.80
Total Capacity	9,701.80 MW
Actual Generation	27,803,425 MWh (27.8 TWh)
Simulated Generation	22,506,880 MWh (22.5 TWh)
Bias	-5,296,545 (-19%)
Average Capacity Factor	26.48%
Farms within $\pm 5\%$	38%
Farms within $\pm 10\%$	47%
Farms with $>20\%$ error	42%

8.3 Best Performing Farms

Farm Name	Capacity	Bias
Green River Wind Farm	194.3 MW	-0.36%
EcoGrove Wind	100.5 MW	-0.37%
Blackstone Wind Farm II	200 MW	+0.49%
Jersey-Atlantic Wind	7.5 MW	+0.56%
Camp Grove Wind	150 MW	-0.56%

8.4 Worst Performing Farms

Thirty seven farms exhibit >50% error with the 2023 calibration. Below is the state concentration of these difficult farms.

State	Number of Problematic Farms	Number of Farms (Total)	Problematic/Total	Percent of Problematic Underestimated
PA	21	25	84%	100%
WV	7	7	100%	100%
MD	5	5	100%	100%
Other States	4	62	6%	100%

The largest 2022 errors are concentrated in PA, WV, and MD, and all of the problematic farms in these states were underestimated. This suggests that the positive terrain based adjustments for PA and WV may not be sufficiently large enough in the general case, and may also be necessary for MD. While the calculations for these plants are very far off the expectation, a small increase in terrain adjustments can lead to pushing a large amount of wind into the cut-in range of a turbine, increasing energy production drastically.

Outside of these 3 states, however, only 4/62 farms faced greater than 50% error, suggesting that this is likely a location/terrain issue.

9. Conclusions and Applications

9.1 Key Achievements

- Developed a high-accuracy calibrated wind generation model with $R^2 = 0.991$ for 2023
- Achieved 68% of farms within $\pm 5\%$ accuracy for 2023
- Showed decent generality with data from 2022 with $R^2 = 0.889$
- Recognized clear adjustments to make the model more applicable for all states / terrain
- Demonstrated NASA POWER data sufficient for high-accuracy modeling
- Created automated, scalable simulation framework

9.2 Limitations

- Certain states still provide a challenge for determining the impact of terrain on wind calculations
- Improvements could be made by calibrating individual states/regions on a larger set of years
- A more formal method for abstracting wind speeds at varying heights may also improve accuracy
- The lack of turbine metadata for each turbine appears to also cause deficit in the prediction, this could be addressed by pinpointing necessary missing data

9.3 Model Applications

This model provides a strong foundation for:

- Grid planning and reliability studies
- Renewable energy integration analysis
- Electricity market forecasting
- Policy analysis and renewable energy targets
- Wind farm performance benchmarking
- Regional wind resource assessment

References

Algorithm Structure

[1] Campbell, Allison M., et al. "A Multi-Decadal Hourly Coincident Wind and Solar Power Production Dataset for the Contiguous United States." Scientific Data, vol. 11, article no. 1121, 2024.

- [2] Millstein, Dev, et al. "A Database of Hourly Wind Speed and Modeled Generation for US Wind Plants Based on Three Meteorological Models." *Scientific Data*, vol. 10, article no. 883, 2023.
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Calibration

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Appendices

Appendix A: Documentation

- NASA POWER: <https://power.larc.nasa.gov/>
- EIA-860: <https://www.eia.gov/electricity/data/eia860/>
- NREL SAM: <https://sam.nrel.gov/>
- PySAM Documentation: <https://nrel-pysam.readthedocs.io/>

Appendix B: Code/Data

GitHub Link: <https://github.com/DukeAidanHall/PJMGenSimulation>

- code/wind_weather.py
- code/run_sam_windpower_nasa.py
- code/compare_generation.py
- README.md
- data/3_2_Wind_Y2022.xlsx
- data/Wind_Turbines.csv